HyperGraph based Imbalance Multi-label Classification with Label Specific Features

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Abstract

The multi-label classification problem involves assigning multiple labels to each instance, which can belong to one or more classes simultaneously. Binary relevance (BR) is a well-known paradigm for multi-label classification but it has a number of drawbacks such as class-imbalance, label inconsistency, and label correlations. While label correlations are often examined in pairwise relationships between instances, real-world systems are often better represented as networks that model complex interactions. In such cases, hypergraphs are more suitable than simple Laplacians. Our proposed approach addresses these issues by applying different weights to positive and negative examples based on class distribution to handle class imbalance, uses label-specific features to tackle inconsistency, and employing hypergraphs to capture complex sample associations. We utilize hinge loss function to reduce sensitivity to outliers and the Accelerated Proximal Gradient (APG) method to efficiently solve the optimization problem. Our approach shows competitive performance compared to existing state-of-the-art methods.

Keywords

Hypergraph, Multi-label learning, Label Specific Feature, accrelated proximal method

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1 Introduction

Multi-label learning is a supervised learning where each instance belongs to one or more than one label simultaneously. It has attracted increasing attention from researchers because of its importance and heavy usage in real life applications such as image annotation [4], [12], video annotation [17], [24], text categorization [21], [42] and bio-informatics [8].

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Many multi-label classifiers face challenges with class imbalance, particularly when dealing with a large number of labels and a low density of positive instances— i.e. when there are far more negative examples than positive ones. This imbalance often leads the classifier to be biased towards the negative class. To address this issue, various research efforts have been made. Authors in [5] have defined four metrics to quantify imbalance present in multi-label datasets. For instance, Zhang et al. [40] introduced a solution involving a binary-class imbalance classifier for each label, alongside several multi-class imbalance learners that work in conjunction with other class labels. The final prediction for each class label is then produced by integrating the outputs from both the binary and multi-class classifiers. To address the class-imbalance problem, Sun et al. [33] proposed a method that combines the synthetic minority oversampling technique with an Ada-Boost support vector machine ensemble and temporal weighting to tackle the class-imbalance problem. Some other recent approaches of multilabel learning are discussed by authors in [18, 19, 26–29].

In [39], the authors explored ensemble-based clustering methods that account for label correlations but did not address the issue of class imbalance. Similarly, [13] focused on the assumption that each label in the original set is linked to a subset of relevant features and incorporated label correlation information. However, this approach also overlooked the problem of class imbalance prevalent in multi-label learning.

The hypergraph-based multi-label learning method leverages the spectral properties of the hypergraph to capture the correlation structure of labels, thereby enhancing the learning process from data. Taking motivation from the aforementioned approaches, we propose to address the class imbalance issue and use the structural information of data via locality of data in order to ensure that similar data instances will have similar class labels. We have also considered label correlation which exist among different possible labels. Finally, we model the multi-label classification problem by selecting label-specific features repeatedly in order to deal with the inconsistency problem. Main contributions of the proposed work can be summarize as follows:

(1) We have considered label-specific data features in order to discriminate each class label.
(2) We apply different weights to positive and negative examples depending on the class distribution.
(3) We impose the structural property of data via incorporating information related to locality of data via HyperGraph. Unlike Laplacian, HyperGraph considers complex relationship that exit in dataset.
(4) We consider correlation that exist among labels.

(5) Comprehensive experimental results show that our proposed model can achieve better or comparable performance than other state-of-the-art methods.

Rest of the paper is organised as follows: Section 2 focuses on related work on multi-label learning. We present our proposed algorithm in Section 3. The proposed algorithm's optimization is discussed in Section 4. Section 5 shows the outcomes of the experiments and evaluations. Concluding remarks are consider in Section 6.

2 Related Work

Multi-label learning approaches can be categorized into three categories: problem transformation approaches, algorithm adaption approaches, and ensemble methods. The problem transformation approaches divide a multi-label problem into one or more binary classification problems which can be solved by single-label classification algorithms such as Naive Bayesian classifier [22], Logistic Regression [23], Support Vector Machines [37], etc.

Algorithm adaption approaches try to bring changes in traditional single-label classification algorithms to use them for multi-label classification problems, for example, ML-kNN [43] is derived by modifying the k-Nearest neighbor algorithm, speeding up k-Nearest neighbors classifier for large-scale multi-label learning on GPUs are discussed in [32]. Ensemble methods use a set of multi-class classifiers for the creation of a multi-label classification problem for example random k-labelsets (RAKEL)[36], that utilizes multiple LP classifiers.

Each class label in multi-label learning can be determined by its own specific characteristics. Multi-label approaches are divided into three categories based on the label-label correlation, namely first-order, second-order, and higher-order methods.

First-order approaches decompose multi-label learning problem into a set of binary classification problems one for each class label without taking into account the label correlation, for example Sparse Weighted Instance-based Multi-label (SWIM) [20], multi-label learning with Label specific features (LIFT)[41], BR [4], and ML-kNN [43], Rank-SVM [16]. First-order based algorithms are mostly simple and efficient but have low accuracy due to ignorance of correlation among labels.

Second-order approaches deal with multi-label learning problems by using the pairwise relationship between label pairs. Using the interaction between label pairs is one of the most popular methods of modeling pairwise relationship, such as Learning Label Specific Features for multi-label classification (LLSF) [13], CPNL [38], calibrated label ranking [10], multi-label teaching-to-learn and learning-to-teach [11], TSMLHN [34], LSLC-ML [6], JFSC [14].

Higher-order approaches to multi-label learning use relationships among all the class labels or mining relationships between a subset of class labels to solve the problem, such as multi-label classification using ensembles of pruned sets (EPS) [30], Classifier Chains (CCs) [31], and Label Powerset (LP) [35].

A hypergraph extends the concept of a traditional graph by allowing edges to be arbitrary non-empty subsets of the vertex set, rather than just pairs of vertices. This generalization has proven effective in capturing complex, high-order relationships across various domains. In our propose approach we use different weights to positive and negative examples depending on the class distribution to tackle with class imbalance issue and apply hypergraph to address the structural property of data named as locality of data to ensure similar instances will have similar class labels and we take advantage of label correlation in our model, finally, we use label-specific features for training the multi-label classification problem.

Given a graph $G = (V, E)$, $V = \{v_1, v_2, ..., v_n\}$ and $E = \{e_1, e_2, ..., e_n\}$ are the vertex and hyper-edges set. We can define a vertex-edge matrix F such that

$$F = f(v, e) = \begin{cases} 1, & v \in e \\ 0, & v \notin e \end{cases} \tag{1}$$

The degree of hyperedge $\delta(e)$ is defined as the number of vertices a hyperedge contains and $d(v)$ represents degree of a vertex,

$$\delta(e) = \sum_{v \in e} f(v, e)$$

$$d(v) = \sum_{v \in e, e \in E} w(e) = \sum_{e \in E} w(e) f(v, e)$$

The following formula defines the hyperedge weight $w(e)$:

$$w(e) = \frac{1}{\delta(e)(\delta(e) - 1)} \sum_{\{v_i, v_j\} \in e} exp\left(-\frac{||v_i - v_j||^2}{\mu}\right) \tag{2}$$

Similar to a simple graph, the Hypergraph's Laplacian matrix can be defined as [44]:

$$H_s = D_v - F D_w D_e^{-1} F^T \tag{3}$$

where D_v, D_e, D_w are the diagonal matrices composed of $d(v), \delta(e)$ and $w(e)$ respectively.

According to Zhou's [44], the Laplacian regularized Hypergraph matrix can also be defined as

$$H_s = I - D_v^{-\frac{1}{2}} F D_w D_e^{-1} F^T D_v^{-\frac{1}{2}} \tag{4}$$

Matrix H_s is a symmetric and positive definite matrix.

3 HyperGraph based Imbalance Multi-label learning using Label Specific Features(HIMLSF)

Notations

Given a multi-label dataset $D = \{(x_k, Y_k)\}_{k=1}^n$ with n instances, the k-th instance is denoted as $x_k \in \mathbf{R}^m$ with its corresponding label vector as $Y_k \in \{-1, 1\}^l$, where l depicts the number of class labels associated with an instance and m stands for the number of features corresponding to an instance. For the sake of convenience, we represent the training data as matrix $X = [x_1, x_2, x_3, ..., x_{n-1}, x_n]^T \in \mathbf{R}^{n \times m}$; $Y = [y_1, y_2, y_3, ..., y_{n-1}, y_n]^T \in \{1, -1\}^{n \times l}$, where k-th instance associated with the q-th label is represented with $y_{kq} = 1$, and $y_{kq} = -1$, if k-th instance is not associated with q-th label. Let the prediction matrix be $F = XW = [f_1, f_2, ..., f_n]^T \in \mathbf{R}^{n \times l}$ and classifier for the multi-label, $H = sign(F)$, where $sign(d)$ returns 1 when $d \geq 0$ and -1 otherwise.

Proposed Formulation

We aim to learn a reasonable prediction model W, based on the latent predicted label vectors. Building on the approach outlined in [29] and aiming to address the challenge of effectively capturing the intricate correlations between labels, we propose creating a hyperedge for each label. This involves grouping all instances that share a common label into a single hyperedge, thereby capturing their collective similarity. To achieve a low-dimensional embedding that preserves the relationships between instances and labels as represented by the hypergraph, we leverage spectral graph embedding theory [7]. The purpose of hypergraph embedding is to find optimal low dimensional representation that maintains the original relationship between the datapoints. Specifically, we use a linear transformation to compute this embedding, ensuring that the instance-label connections are maintained.

Based on the Laplacian of the constructed hypergraph, we propose the optimization problem defined as follows:

$$\min_{W} \frac{1}{2} ||| \, (|E - Y \circ (XW)|_{+})^2 \circ M\|_1 + \lambda_1 ||W||_1 + \lambda_2 Tr(WRW^T) + \lambda_3 Tr((XW)^T H_s XW). \tag{5}$$

where $W \in \mathbf{R}^{m \times l}$ is the regression coefficient, E denotes the corresponding matrix where each element equals 1 and $\circ$ denotes the Hadamard (element-wise) product of matrices, $\lambda_1 \geq 0$ is a regularization parameter. $M \in \mathbf{R}^{n \times l}$ handles the class imbalance issue and indicates the weighted matrix for all n instances corresponding to l classes, and its values are calculated as follows:

$$m_{kq} = \begin{cases} \left(\frac{\sum_{j=1}^{n} y_{jq} == -1}{\sum_{j=1}^{n} y_{jq} == 1} \right)^{\beta} & if \;\; y_{kq} = 1 \\ 1 & otherwise, \end{cases} \tag{6}$$

where $\beta \in [0, 1]$ regulates the extent of a weighting factor and if $\frac{\sum_{j=1}^{n} y_{jq} == -1}{\sum_{j=1}^{n} y_{jq} == 1} = 5$, it shows that in a given class label q, the number of negative instances is five times greater than the number of positive instances, and the weight for positive instances is $5^{\beta} \in [1, 5]$ based on the chosen β value, while the weight for negative instances is always 1.

The first term in our proposed model (equation 5) is measuring the risk between the ground-truth label and the prediction value using least squared hinge loss function $loss(y, g(d)) = (max(0, 1 - yg(d)))^2 = (|1 - yg(d)|_{+})^2$ instead of least squared loss function [38], which is more suitable for the multi-label classification problem and robust against outliers.

The second term deals with selecting the label-specific feature with the assumption that each class label could be determined by some specific features of its own. We used L_1 regularization to select the label-specific features corresponding to each label because the L_1 norm is used to make our weight matrix sparse [13], which allows us to keep the important features for a specific label while removing the features that aren't important for a specific class label. Mainly, non-zero values of each $W_k \in \mathbf{R}^m$ corresponds to label-specific feature.

The third term in our proposed model considers that if two labels are highly correlated (i.e., C_{kq} or C_{qk} is large), they will share label-specific features. As a result, the corresponding model coefficients W_k and W_q will be very close, with a small Euclidean distance between them. Otherwise, W_k and W_q would be dissimilar, resulting in a large Euclidean distance between them. Here $R \in \mathbf{R}^{l \times l}$, $R_{kq} = 1 - C_{kq}$ and C_{kq} indicates the correlation coefficient between two labels y_k and y_q, which we calculate using cosine similarity in our propose model.

In order to incorporate concept of locality of data (i.e. similar pair of instances must have similar predicted label vectors, we use the Laplacian Hypergraph (H_s) on a scatter of data points for modeling the geometric structure of data.

4 Solution and final problem

The optimization problem (5) is convex, but non-smooth due to the existence of L_1 norm regularization term. To solve this non-smooth optimization problem, we're using accelerated proximal gradient descent. Thus, we can write the optimization problem as:

$$\min_{W} G(W) = g(W) + f(W), \tag{7}$$

where

$$g(W) = \frac{1}{2} ||| \, (|E - Y \circ (XW)|_{+})^2 \circ M\|_1 + \lambda_2 Tr(WRW^T)^T) + \lambda_3 Tr((XW)^T H_s XW), \tag{8}$$

$$f(W) = \lambda_1 ||W||_1. \tag{9}$$

Note, although $g(W)$ and $f(W)$ are convex functions, but the latter is not smooth. Working on the lines of [3], we are minimizing $G(W)$ via sequence of separable quadratic approximation to $G(W)$, i.e. $Q_L(W, W^k)$, we get

$$Q_{L_f}(W, W^k) = g(W^k) + \langle \nabla g(W^k), W - W^k \rangle + \frac{L_f}{2} ||W - W^k||_F^2 + f(W). \tag{10}$$

Here, $Q_{L_f}(W, W^k) \geq G(W)$ will always be found, for any $L \geq L_f$, where L_f denotes the Lipschitz constant. Following that, the proximal algorithm minimises a set of separable quadratic equations to $G(W)$. We can solve W by minimizing $Q_L(W)$ as in the paper [3], the following is the formula:

$$\begin{aligned} W^* &= arg \min_{W} Q_L(W, W^i), \\ &= arg \min_{W} (f(W) + \frac{L}{2} ||W - (W^i - \frac{1}{L} \nabla g(W^i))||_F^2), \end{aligned} \tag{11}$$

We can improve the convergence rate to $O(\frac{1}{t^2})$ by setting $W^{(t)} = W_t + \frac{\beta_{t-1}}{\beta_t}(W_t - W_{t-1})$ as discussed in [2], in which the series β_t satisfies $\beta_{t+1}^2 - \beta_{t+1} \leq \beta_t^2$, where W_t represents the result of W at the t^{th} iteration.

4.1 Update W

For updating W , we need to calculate the derivative of $g(W)$ concerning W as follows:

$$\nabla_W g(W) = X^T \; (|E - Y \circ (XW)|_{+} \circ (-Y) \circ M) + \lambda_2 WR + \lambda_3 X^T H_s XW. \tag{12}$$

Now, we can update W as determined by Eq.(11),

$$W^t = W_t + \frac{\beta_{t-1} - 1}{\beta_t}(W_t - W_{t-1}), \tag{13}$$

$$W^{t+1} = prox_{\varepsilon}(W^t - \frac{1}{L}\nabla_W g(W^t)), \tag{14}$$

where ε is the step size. Here, since W is $L_1 - norm$ regularization term in $f(W)$, we need to solve this value using the element-wise soft–threshold operator as follows:

$$prox_{\varepsilon}(W_{kq}) = (|W_{kq}| - \varepsilon)_{+} sign(W_{kq}). \tag{15}$$

4.2 Proof of Lipschitz continuity

We will prove Lipschitz continuity of $g(W)$. Thus, for given W_1 and W_2, we have

$$\begin{aligned}
&\|\nabla g(W_1) - \nabla g(W_2)\|_F \\
&= \Big\| X^T\big(|E - Y \circ XW_1|_{+} - |E - Y \circ XW_2|_{+}\big) \circ (-Y) \circ M \\
&\quad + \lambda_2 \Delta W R + \lambda_3 X^T H_s X \Delta W \Big\|_F \\
&\le 3\|X^T\big(|E - Y \circ XW_1|_{+} - |E - Y \circ XW_2|_{+}\big) \circ (-Y) \circ M\|_F^2 \\
&\qquad + 3\|\lambda_2 R \Delta W\|_F^2 + 3\|\lambda_3 X^T H_s X \Delta W\|_F^2 \\
&\le 3\max(m_{kq})^2 \|X^T\|_F^2 \||E - Y \circ XW_1|_{+} - |E - Y \circ XW_2|_{+}\|_F^2 \\
&\qquad + 3\|\lambda_2 R\|_F^2 \|\Delta W\|_F^2 + 3\|\lambda_3 X^T H_s X\|_F^2 \|\Delta W\|_F^2 \\
&\le 3\max(m_{kq})^2 \|X^T\|_F^2 \|X^T\|_F^2 \|\Delta W\|_F^2 + 3\|\lambda_2 R\|_F^2 \|\Delta W\|_F^2 \\
&\qquad + 3\|\lambda_3 X^T H_s X\|_F^2 \|\Delta W\|_F^2 \\
&\le 3\max(m_{kq})^2 \big(\|X^T\|_F^2\big)^2 \|\Delta W\|_F^2 + 3\|\lambda_2 R\|_F^2 \|\Delta W\|_F^2 \\
&\quad + 3\|\lambda_3 X^T H_s X\|_F^2 \|\Delta W\|_F^2
\end{aligned}$$

As a result, the Lipschitz constant would be,

$$L_f = \sqrt{\big(3\max(m_{kq})^2 \|X^T\|_F^2\big)^2 + 3\|\lambda_2 R\|_F^2 + 3\|\lambda_3 X^T H_s X\|_F^2}. \tag{16}$$

5 Experimental Section

The experiments are performed using five-fold cross validation in MATLAB version 2020a under Microsoft Windows environment on a machine with 3.40 GHz CPU and 16 GB RAM on six well known multi-label datasets, which are taken from the site http://mulan.sourceforge.net/datasets-mlc.html.

5.1 Compared Algorithms

We have compare our propose algorithm with the following state-of-the-art for the multi-label learning algorithm. We have use all configurations for parameters as it was suggested in the original paper.

(1) CPNL[38] It is a second-order strategy that applies a cost-sensitive multi-label learning model of pairwise positive and negative label pairwise correlation, as well as addressing the issue of class imbalance by computing the cost sensitive loss matrix and incorporating it into the loss function. The parameter $\beta = 0.5$, $k = 3$ and the hyper-parameters λ_1, λ_2, λ_3 are tuned in $\{10^{-4}, 10^{-3}, \ldots, 10^{2}\}$.

(2) JFSC [14]It uses pair-wise label correlation to learn label-specific and shared features, and it additionally solves the class imbalance issue using a fisher discriminant-based regularization term and the parameters γ,β and α are tuned in $\{4^{-5}, 4^{-4}, \ldots, 4^{5}\}$, and η is tuned in {0.1, 1, 10}.

Algorithm 1 HIMLSF Optimization

Require: $X \in R^{n \times m}$, $Y \in \{-1, 1\}^{n \times l}$
Ensure: W;
Initialization; Model Parameters W; Trade-off parameters: $\lambda_1; \lambda_2; \lambda_3$;
Iteration parameters: $\beta_1; \beta_2; t$;
Relative parameters: $L_f; L$;
Calculate the Laplacian HyperGraph matrix H_s;
Calculate R via C matrix, where C is calculated using cosine similarity;
Calculate R via $R_{kq} = 1 - C_{kq}$, where C_{kq} is the correlation coefficient between two labels y_k and y_q;
Repeat
while G(W) > $Q_L(W, W^t)$ **do**
    $L = \eta L$ where η is the search step size;
end while
Calculate W^t by (13);
$G_w^t \leftarrow W^t - \frac{1}{L}\nabla_W g(W^t)$
$W_{t+1} \leftarrow prox_{\frac{\lambda_1}{L}}(G_w^t)$; by (15)
$\beta_{t+1} \leftarrow \frac{1+\sqrt{4\beta_t^2+1}}{2}$
$t \leftarrow t + 1$
Until MaxIteration Reached
$W_* = W_t$
Return W^*

(3) FSRD [25] ranking-based feature selection algorithm for multi-label classification that addresses the class imbalance issue by calculating the fuzzy relative discernibility relation on each object pair and then employing the MLKNN algorithm for multi-label classification.

(4) S-CLS [1] It employs a unified structure based on the Laplacian score for semi-supervised multi-label function collection and then uses the MLKNN algorithm for multi-label classification.

(5) MLKNN [43] It works based on the traditional KNN algorithm and introduces maximum posterior probability (MAP) for dealing with multi-label classification problems. The value for parameter k is in $\{3, 5, \ldots, 21\}$.

(6) LPLC [15] The local positive and negative pairwise label correlations are explored base on the traditional KNN algorithm. The value for parameter k is tuned in $\{3, 5, \ldots, 21\}$ and the tradeoff parameter α is tuned in $\{0.1, 0.2, \ldots, 1\}$.

(7) LLSF [13] It exploits the concept of learning label-specific features for training the multi-label classification model. The parameters λ is tuned in $\{2^{-10}, \ldots, 2^{10}\}$.

(8) IMLSF [29] It addresses class imbalance by applying different weights to labeled and unlabeled instances of a given class based on the class distribution. The parameters λ is tuned in $\{2^{-10}, \ldots, 2^{10}\}$.

(9) HIMLSF We address class imbalance by applying different weights to labeled and unlabeled instances of a given class based on the class distribution and also Laplacian Hyper-Graph to address higher order relationship present in multi-label data.The multi-label classification model is then trained

using the principle of learning label-specific features.The parameters λ is tuned in $\{2^{-10}, ..., 2^{10}\}$.

5.2 Computational Complexity

In this section, we will briefly examine the optimization problem's time complexity. For matrix $X \in R^{n\times m}$, $Y \in R^{n\times l}$, $W \in R^{m\times l}$, where n is the total number of examples, the total number of labels for corresponding instances is indicated by l, and the number of features for corresponding instances is indicated by m. In our proposed model, the time complexity needed to calculate matrix C is $O(l^2)$, the time complexity to derive Lipschitz constant is $O(l^2{+}mn^2{+}nm^2)$, to derive matrix W, we need $O(mnl + l^2 + ml^2 + n^2 l)$.

Table 1: The statistical information of data sets.

Data set	#Instances	#Features	#Labels	Card	Domain
Birds	645	260	19	1.014	Audio
CAL500	502	68	174	26.044	Music
Emotions	593	72	6	1.868	Music
Flags	194	19	7	3.392	Image
Image	2000	294	5	1.236	Image
Medical	978	1449	45	1.245	Text

5.3 Datasets

The characteristics of six real-world multi-label data sets used are discuss in Table 1. Cardinality (Card) measures the average number of labels associated with each instance.

5.4 Evaluation Metrics

We have evaluated the performance of the compared algorithms in terms of five common metrics namely Hamming loss, Micro F1, Ranking loss, Average Precision, AUC. Given a test data set $T_d = \{x_k, Y_k\}_{k=1}^{N_d}$, where $Y_k \in Y$ is the set of ground truth labels, and $h(x_k)$ is the set of predicted labels for the k^{th} instance. Let $f(x_k, y)$ indicates the confidence score of a point x_k belonging to the label y.

(1) Hamming Loss(HL): It determines the number of times an instance-label pair has been misclassified, i.e., a label associated with an instance is not predicted or a label not associated with an instance is predicted.

$$\text{HLoss} = \frac{1}{N_d}\sum_{k=1}^{N_d}\frac{1}{l}|h(x_k)\Delta Y_k|,$$

where the symmetric difference between two sets is indicted by Δ, N_d indicates the number of instances and l indicates the number of labels.

(2) Micro F1 (F1): It is a more advanced version of the F1 Measure that treats each label vector entry as a single example.

$$\text{MicroF1} = \frac{2\sum_{q=1}^{l}\sum_{k=1}^{N_d} y_{kq}h(x_{kq})}{\sum_{q=1}^{l}\sum_{k=1}^{N_d} y_{kq} + \sum_{q=1}^{l}\sum_{k=1}^{N_d} h(x_{kq})}.$$

(3) Ranking Loss (RL): This determines the percentage that the ranking of negative labels of the example are higher than that of positive labels.

$$\text{RL} = \frac{1}{N_d}\sum_{k=1}^{N_d}\Big(\frac{1}{|Y_k||\bar{Y_k}|}|\{(y', y'')|f(x_k, y') \leq f(x_k, y'')| \quad (y', y'') \in Y_k \times \bar{Y_k})\}|\Big).$$

(4) Average Precision(AP): This evaluates the average fraction of positive labels that are higher than the particular labels.

$$\text{AP} = \frac{1}{N_d}\sum_{k=1}^{N_d}\frac{1}{|Y_k|}\sum_{y\in Y_k}\frac{|\{y'|\text{rank}_{f(x_k,y')} \leq \text{rank}_{f(x_k,y)}, y' \in Y_k\}|}{\text{rank}_{f(x_k,y)}}.$$

(5) Average AUC(AUC): Calculates the average fraction of times a positive instance of all-class labels has a higher ranking than a negative instance.

$$\text{AUC} = \frac{1}{l}\sum_{k=1}^{l}\frac{|\{(x', x'')|f(x', y_q) \geq f(x'', y_q)|(x', x'') \in U_q \times \bar{U_q}\}|}{|U_q||\bar{U_q}|},$$

where $U_q = \{x_k|y_q \in Y_k, 1 \leq k \leq l\}$ ($\bar{U_q} = \{x_k|y_q \notin Y_k, 1 \leq k \leq l\}$)denotes the set of test instances with(without) label y_q.

5.5 Results and Statistical Analysis

The average results on six data sets, in terms of five evaluation metrics, for all the compared algorithms are reported in Table 2. In order to discuss the impact of the regularization parameters we have compared the performance of the proposed algorithm, HIMLSF, on Cal500 dataset in Fig 1.

We further analyze pairwise comparisons using the Nemenyi test [9]. For eight classifiers, the critical value, $\alpha_{0.5}$ is 2.850 and the corresponding CD is $\alpha_{0.5}\sqrt{\frac{k(k+1)}{6N}}$ i.e. 1.8851, where k is the number of competing classifiers and N represents the number of data points. Significant differences exist between the best and worst-performing algorithms across all metrics except Hamming loss, as shown in Figures 2 (Algorithms that are not connected with the horizontal lines are significantly different). Figure 3 shows the consolidated avg. rank across all metrics. Overall, HIMLSF demonstrates competitive and on average superior performance.

The experimental findings lead to several key observations. LPLC focuses solely on the local correlations between positive and negative labels, while LLSF incorporates both label-specific features and label correlations. FSRD and S-CLS perform feature selection before applying the MLKNN algorithm for classification, but these methods also face challenges due to label imbalance in the training dataset. Unlike these approaches, MLkNN does not leverage any potential associations among labels. On the other hand, JFSC and CPNL address class imbalance and consider label associations, but neither enforces structural sparsity in prediction error, which could enhance robustness against corrupted features. Although JFSC and CPNL recognize label-specific features in multi-label learning, they do not fully address the problem of inconsistency, as CPNL does not. Our proposed algorithm tackles both class imbalance and imposes a structural property known as data locality, ensuring that similar data instances have similar class labels. This approach significantly

Table 2: Experimental results of eight comparing algorithms in terms of five evaluation metrics. ↑(↓) denotes the higher (smaller) the value, the better the performance. The most promising results are bolded.

Dataset	Metric	LLSF	JFSC	FSRD	S-CLC	MLKNN	LPLC	CPNL	IMLSF	HIMLSF
Cal500	HL(↓)	0.1368	0.1368	0.138	0.1382	0.1383	0.1692	0.1407	0.1359	**0.1358**
	F1(↑)	0.3144	0.3142	0.3125	0.3163	0.314	**0.4212**	0.2305	0.3393	0.3353
	AP(↑)	0.5037	0.5076	0.4904	0.4924	0.4931	0.4552	0.4952	0.5169	**0.5179**
	RL(↓)	0.1801	0.1789	0.1842	0.1828	0.1837	0.2332	0.1872	0.1741	**0.1738**
	AUC(↑)	0.8177	0.818	0.8121	0.8137	0.8125	0.771	0.8096	0.823	**0.8231**
Emotions	HL(↓)	0.2999	0.2903	0.2636	0.2686	0.2634	0.3	0.2496	0.201	**0.1956**
	F1(↑)	0.1358	0.2116	0.4733	0.4595	0.4893	0.546	0.5041	0.623	**0.6264**
	AP(↑)	0.6391	0.6906	0.6989	0.6928	0.7054	0.6903	0.7252	0.8073	**0.8186**
	RL(↓)	0.3261	0.2715	0.2682	0.2724	0.2605	0.2918	0.2424	0.1541	**0.1469**
	AUC(↑)	0.6435	0.6983	0.7057	0.7028	0.7126	0.69	0.7309	0.8168	**0.8240**
Birds	HL(↓)	0.0534	0.0535	0.0543	0.0541	0.054	0.0606	0.0623	0.043	**0.0419**
	F1(↑)	0	0	0.0433	0.0113	0.0335	0.1831	0.2377	0.3506	**0.3598**
	AP(↑)	0.236	0.2514	0.4187	0.4063	0.4261	0.2051	0.4291	0.7773	**0.7881**
	RL(↓)	0.1485	0.1413	0.2764	0.2877	0.2788	0.196	0.2759	0.1028	**0.1013**
	AUC(↑)	0.3761	0.3803	0.3739	0.3706	0.3723	0.3439	0.3722	0.5294	**0.5307**
Flags	HL(↓)	0.2983	0.2918	0.3293	0.3233	0.3093	0.3366	0.2936	0.2865	**0.2686**
	F1(↑)	0.6661	0.6678	0.6531	0.6574	0.6647	0.6809	0.6841	0.6694	**0.6876**
	AP(↑)	0.8018	**0.8049**	0.8024	0.8016	0.8007	0.7834	0.7958	0.7946	0.7994
	RL(↓)	**0.2281**	0.2233	0.2334	0.2289	0.2323	0.2612	0.24	0.2409	0.2395
	AUC(↑)	0.7291	**0.734**	0.7283	0.7295	0.7278	0.7084	0.7188	0.7195	0.7237
Medical	HL(↓)	0.0103	0.0101	0.0165	0.0156	0.0157	0.0175	0.0105	0.0099	**0.0098**
	F1(↑)	0.8069	0.8107	0.6579	0.6844	0.6797	0.6579	0.8115	0.8185	**0.8199**
	AP(↑)	0.9064	0.8983	0.803	0.8045	0.806	0.7702	0.9109	0.91	**0.9114**
	RL(↓)	0.0226	0.0222	0.0441	0.0464	0.0464	0.0621	**0.0118**	0.016	0.0165
	AUC(↑)	0.9739	0.9732	0.9486	0.9461	0.9456	0.932	**0.9835**	0.9799	0.9781
Image	HL(↓)	0.1879	0.1841	0.1784	0.1757	0.1789	0.5896	0.1957	0.1786	**0.1777**
	F1(↑)	0.4737	0.5126	0.5334	0.5318	0.5381	0.5056	**0.6121**	0.6059	0.6077
	AP(↑)	0.7874	0.7866	0.7732	0.7726	0.7789	0.694	0.7862	0.788	**0.7949**
	RL(↓)	0.177	0.1789	0.195	0.1927	0.1883	0.28	0.177	0.1746	**0.1679**
	AUC(↑)	0.7812	0.7793	0.7635	0.7659	0.7709	0.7002	0.7813	0.7833	**0.79**

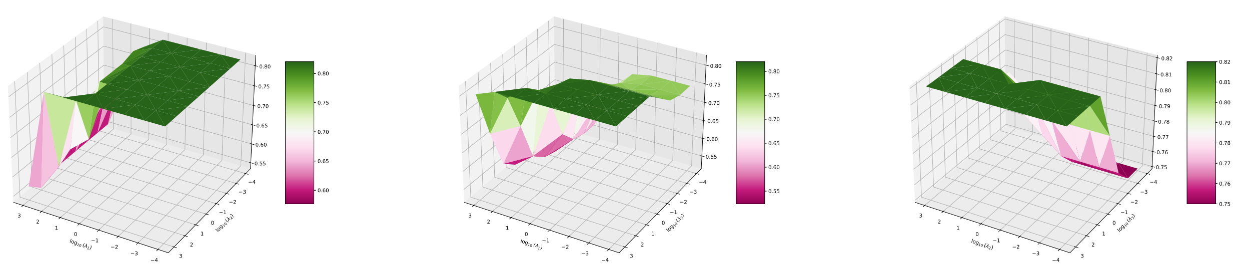

Figure 1: Ablation analysis of HIMLSF

improves classifier performance. Comparisons across various scenarios illustrate the superior effectiveness of our algorithm relative to other state-of-the-art multi-label learning methods discussed in this paper.

6 Conclusions

In this paper, we introduce a novel algorithm called HyperGraph-based Imbalance Multi-Label Data Learning with Label-Specific Features (HIMLSF). HIMLSF tackles the challenge of class imbalance by assigning distinct weights to positive and negative instances

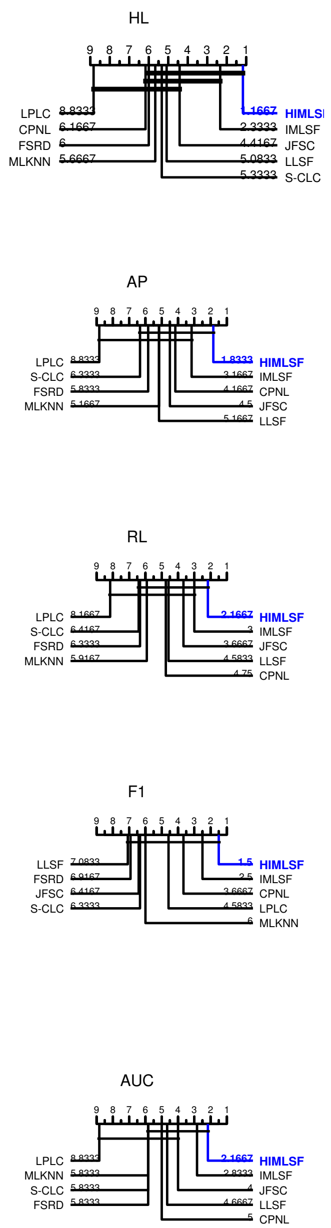

Figure 2: HIMLSF comparison with competing algorithms using Nemenyi test.

of each class label. Additionally, it leverages label correlations using hypergraph to boost classifier accuracy and incorporates a structural property known as data locality to ensure that similar instances receive similar labels. The algorithm also includes a feature selection component that focuses on label-specific features. To efficiently and effectively solve the objective function, we employ the Accelerated Proximal Gradient method, which allows for

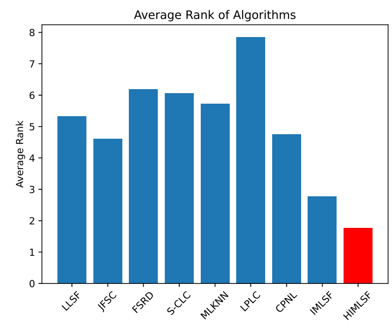

Figure 3: Consolidated average rank of participating algorithms across all metrics.

rapid iterative computation. Experimental results demonstrate that HIMLSF significantly outperforms existing methods in both multi-label classification and feature selection tasks. A future line work of is to scale the algorithm for extreme multi-label scenario.

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